

Mayank Kumar Singh

Full Stack Java Developer — GenAI Enthusiast

[GitHub](#) — [LinkedIn](#) — mayankkumarsingh262004@gmail.com

SKILLS

Languages: Java, Python, C, JavaScript, HTML, CSS
Backend: Spring Boot, REST APIs, Microservices (Basic), Hibernate/JPA
Databases: MySQL, PostgreSQL, MongoDB
Tools: Git, Docker, Maven, Postman, Linux
AI/GenAI: OpenAI API, Gemini API, Ollama (Local LLM), Prompt Engineering
Concepts: OOP, Multithreading, DSA, DBMS, OS, Computer Networks, MVC Architecture

COURSEWORK

Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems

PROJECTS

AI Resume Analyzer (Deployed on AWS)

[Live](#) — [GitHub](#)

- Developed a full-stack AI-powered resume analysis platform using Spring Boot, PostgreSQL, and JavaScript.
- Implemented JWT-based authentication for secure user access and session management.
- Integrated Gemini API to analyze resumes and generate match scores, skill insights, and improvement suggestions.
- Built features including resume upload, real-time analysis dashboard, and user-specific history tracking.
- Containerized the application using Docker and deployed it on AWS EC2 with proper environment configuration.
- Configured networking, security groups, and backend connectivity for production-level deployment.

Brain Tumor Detection System with AI Chatbot

[GitHub](#)

- Built an AI-based web application for brain tumor detection using MRI images and machine learning models.
- Integrated Python-based ML model with Spring Boot backend for real-time prediction via REST APIs.
- Implemented an AI-powered chatbot using Ollama (local LLM), enabling offline inference and enhanced user interaction.
- Designed scalable backend services focusing on performance, API efficiency, and real-time data processing.

AI Chatbot using Spring Boot and Ollama

[GitHub](#)

- Developed an AI chatbot using Spring Boot and Ollama for local LLM-based response generation.
- Implemented REST APIs for handling user queries and delivering real-time conversational responses.
- Optimized response latency and improved user interaction experience.

URL Shortener Service using Spring Boot

[GitHub](#)

- Designed scalable URL shortening service with unique ID generation and redirection system.
- Implemented REST APIs and optimized database queries for high performance.
- Optimized database queries and ensured high availability for fast URL redirection.

EDUCATION

NITRA Technical Campus

Bachelor of Technology (B.Tech)

Final Year (8th Semester)

ACHIEVEMENTS

- Solved 100+ DSA problems on LeetCode focusing on problem-solving, optimization, and coding efficiency.

ADDITIONAL

- Hands-on experience with AWS EC2 deployment, Docker containerization, and cloud-based application hosting.
- Familiar with Linux environment and command-line operations.
- Strong understanding of system performance, scalability, and reliability.